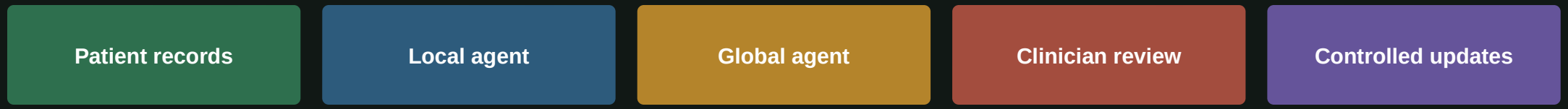
HEALTHCARE CASE STUDY

Dagents for Source-First Hospital Warnings

Start with sepsis, then reuse the source-first coordination layer for other high-impact conditions—without pretending one model fits every disease.



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Professor presentation

WHERE THIS PATTERN HELPS

Regulated work needs local control and shared coordination.

Healthcare and finance both have scattered data, strict access rules, changing models, and heavy evidence requirements. Healthcare starts with sepsis and expands condition by condition.

- Healthcare: sepsis, stroke, heart attack, sudden kidney injury, serious breathing decline, repeat visits, and long-term care.
- Finance: fraud, money laundering, changing loan risk, cash-flow risk, and unusual transactions.
- Why sepsis first: it tests urgent, local, multi-hospital coordination; it is not the platform's permanent disease scope.
- Scope: test past records, then silent live runs—never diagnose or treat automatically.

FIRST CASE SEPSIS

approved summary

GLOBAL COMPARISON

WHY SEPSIS

Sepsis is a strong first test because delays matter.

Sepsis is the body's extreme response to infection. It can damage organs quickly, so hospitals need timely and dependable warning support.

48.9 million cases

WHO summary of a published global estimate.

11 million deaths

About one in five deaths in that global estimate.

15 per 1,000

Estimated hospitalized patients who develop sepsis during care.

Why it is difficult

Warning signs are spread across patient records, lab results, devices, and hospital routines.

Evidence: WHO Sepsis fact sheet, 3 May 2024

WHY START WITH SEPSIS?

Start with sepsis; add other conditions one by one.

Dagents can reuse the coordination platform, but every condition still needs its own tested model, inputs, workflow, and safety evidence.

Time matters

A delayed warning can matter, so the system must be fast, reliable, and easy to stop safely.

Many routine signals

The warning may use vital signs, lab results, medicines, and visit information already found in hospitals.

Hospitals differ

Patient populations, data formats, workflows, and warning limits vary—exactly what local and global agents must handle.

Safe staged testing

Teams can test past data, then run silently, then try a limited pilot before affecting care.

OTHER CONDITIONS

Sepsis is the first case—not the only one.

The same source-first pattern can support other urgent hospital conditions, with a separately tested model and workflow for each.

Stroke

Combine symptoms, vital signs, time information, and the imaging workflow to support fast review.

Heart attack

Use symptoms, heart tracing, blood tests, and vital signs to support quick local review.

Sudden kidney injury

Watch changes in kidney-related tests, urine output, medicines, and blood pressure over time.

Breathing decline

Watch oxygen, breathing rate, blood tests, devices, and response to treatment for worsening.

THE CORE PROBLEM

The hard part is coordination—not just prediction.

A useful warning system must handle different data, different hospital rules, safe updates, and clear evidence of what happened.

Scattered data

Values arrive in different formats, units, and times. Dagents checks and standardizes them locally.

Different hospitals

Patients, workflows, and acceptable warning volume vary. Dagents keeps local settings while comparing approved results.

Repeated platform work

Without a shared framework, every site rebuilds scheduling, model runs, deployment, monitoring, and rollback.

Late audit evidence

Approvals, changes, overrides, and run history are often gathered after the fact. Dagents records them as work happens.

Result: duplicated engineering, slower rollout, uneven behavior, and weak cross-hospital learning / 06

WHAT THE SYSTEM MUST DO

The case needs a source-first system—not a central data copy.

Five capabilities turn a model into controlled warning support while detailed records stay with each hospital.

Connect beside the source

Read approved hospital data where it already lives, standardize units, check missing or old values, and record where each value came from.

Accumulate + run locally

The local agent keeps source-side history, calculates risk inside the hospital, blocks unsafe runs, and records checks, versions, warnings, and outcomes.

Help the clinician

Show why the warning appeared, how certain it is, and known limits. Record the clinician's response.

Coordinate with small summaries

Approved counts, summaries, versions, and pointers leave—not raw patient rows.

Update safely

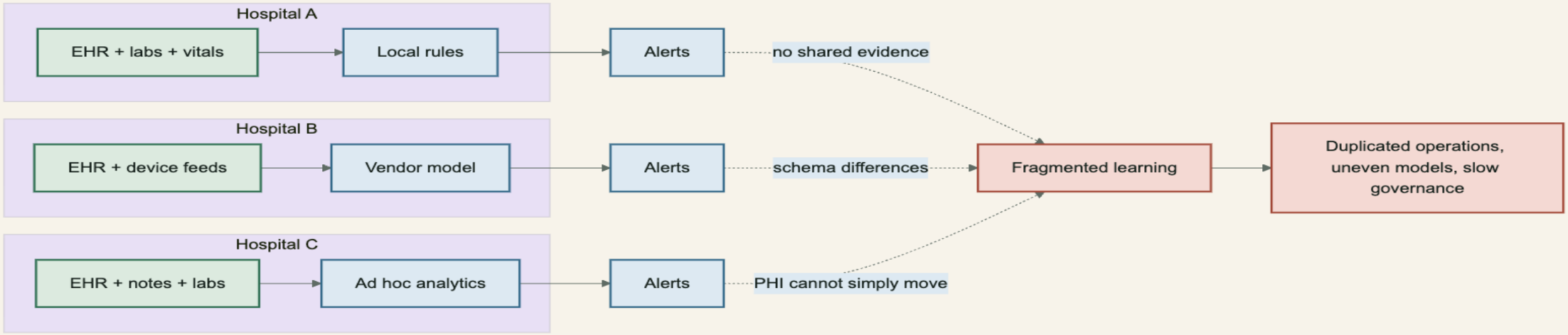
Track versions, test small rollouts, monitor results, and return to an older version if needed.

Controls across every step: minimum data movement, security, privacy, change history, local retention, and clear human ownership.

CURRENT STATE

Separate hospital systems make learning difficult.

A common response is to copy everything into one warehouse or lake—but that adds delay, repeated data movement, and another sensitive copy.



WHY NOT A CENTRAL DATA LAKE?

Centralizing raw records adds delay, volume, and risk.

For time-sensitive hospital warnings, moving everything first is unnecessary.

Slower response

Extract, transfer, clean, and load must finish before central analysis. New data and model feedback arrive later.

Large recurring transfers

Detailed records move repeatedly, use network capacity, and require expensive ingestion pipelines.

More copies, larger exposure

A central copy adds access, breach, retention, and deletion risk across hospitals.

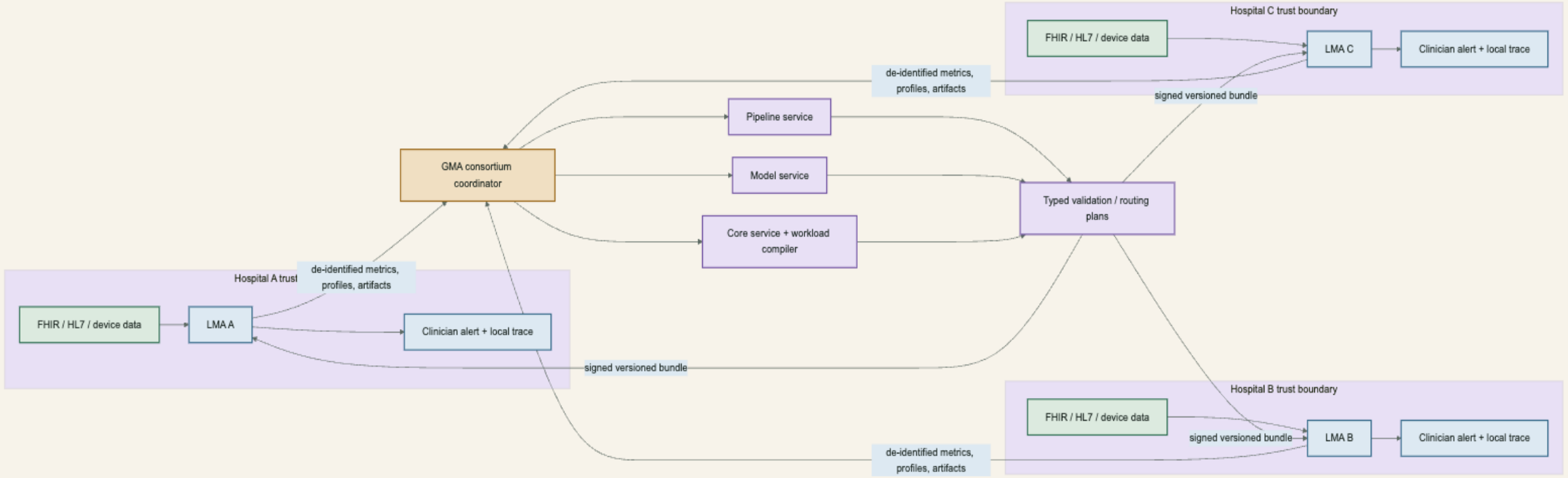
Constant integration work

Every source or schema change forces updates to central ingestion and normalization.

SOURCE-FIRST DESIGN

Dagents keeps data and computation at the source.

Hospitals keep detailed records and run history; the global agent receives only small approved coordination data.



LOCAL AGENT (LMA)

The local agent builds the useful history at the source.

It connects to local systems, runs beside them, and keeps checks, versions, warnings, responses, and outcomes under hospital rules.

READ LOCALLY

Read approved patient-record data and device readings inside the hospital—without first copying them to a central store.

CHECK + RECORD

Check format, units, age, allowed use, and missing values; keep source-side run history.

CALCULATE LOCALLY

Run the approved model beside the source; return a risk score, main reasons, and model version.

SHOW

Show the warning to a clinician and record the response and outcome.

Only approved counts, performance summaries, exceptions, versions, and pointers may leave; raw patient rows do not.

GLOBAL AGENT (GMA)

Global coordination uses small summaries.

No central patient-data lake is required.

REGISTER

Track each hospital, local agent, version, capability, and sharing rule.

COLLECT MINIMUM

Receive counts, quality and performance measures, run summaries, exceptions, versions, and pointers—not raw patient rows.

COMPARE SUMMARIES

Compare warning quality, changes over time, workload, and reliability using the approved small payload.

PLAN + UPDATE

Choose settings for each hospital, test with a small rollout, monitor, pause, or return to an older version.

The global agent coordinates learning and updates; it is not a patient-record warehouse and does not make care decisions.

ONE RUN

One patient warning becomes a safe learning signal.

Every run follows the same simple path, while the final care decision remains human.

DATA ARRIVES

New vital signs, lab results, medicines, and visit information reach the local agent.

CHECK

The local agent checks permission and data quality, then runs the approved model.

REVIEW

The clinician sees the warning and its main reasons, then decides whether to act.

SHARE

A small approved summary is shared so hospitals can learn from each other without moving the patient record.

If a check fails, no warning is shown; the reason stays in local history and only a small operational exception is shared.

COORDINATION LAYER

Dagents coordinates while patient data stays local.

It registers agents, checks local summaries, runs models at the source, and sends versioned update plans.

REGISTER

Record each hospital and local agent, what it can do, and which version it should run.

CHECK

Compare small local summaries without copying patient records.

RUN

Send a clear model request to the local agent; patient-level execution stays inside the hospital.

UPDATE

Send versioned plans, watch health, and return to a safe version—using small control messages.

Still needed for production: durable storage, reliable messaging, signed updates, and tamper-resistant logs.

COMPLIANCE

Compliance needs more than architecture.

The framework can enforce repeatable technical checks. Hospitals still own policies, staff training, security operations, privacy decisions, and clinical safety.

HIPAA

U.S. rules for protecting patient health information. Dagents can keep identifying data and most evidence at the source, minimize transfers, and record runs. Hospitals still need access controls, encryption, retention rules, risk reviews, and incident response.

SOC 2

An independent review of service security and reliability controls. Dagents can provide health checks and change records. The organization still needs control owners, access reviews, vendor checks, incident evidence, and an auditor.

CLINICAL AI SAFETY

Rules and evidence for software that informs care. Dagents can track model versions, reasons, approvals, and rollback. Teams still need a clear intended use, clinical testing, usability work, and safety review.

Source-first design reduces data movement and copies; real compliance still depends on controls being implemented, used, tested, and reviewed / 15

WHO OWNS WHAT?

Dagents helps; hospitals stay accountable.

Each requirement needs both a technical mechanism and a named person or team responsible for it.

Area	Dagents helps with	Hospital team still owns
Patient-data security	Keep work and run evidence local; limit data sources; record versions and runs	User access, login protection, encryption, backups, audits, and incident response
Patient privacy	Share only approved small summaries and pointers; no raw-patient lake	Purpose, permission, minimum data, retention, deletion, and legal decisions
Service assurance	Repeatable deployment, health checks, and change history	Control owners, access and vendor reviews, operating proof, and external audit
Clinical safety	Human decision point, model and input history, approvals, and rollback	Clinical testing, usability, intended use, regulation, and safety ownership
Health-data standards	A clear connector boundary for hospital data	Correct coding, terminology, source tracking, and patient matching

SAFETY CHECKS

Four checks must pass before a warning is shown.

Any failed check stops the run, records the reason, and sends the issue for review.

1. APPROVED USE

Is this data allowed for this patient group and this purpose?

2. GOOD DATA

Is the data recent, complete enough, correctly formatted, and in the right units?

3. MODEL

Is the correct tested model and hospital setting being used?

4. CLINICIAN

Can the clinician see the reasons and limits and make the final care decision?

Start with past data, then run silently. Stop safely and record why whenever a check fails.

EFFORT

Dagents reduces repeated platform work.

The benefit grows when more hospitals, models, or regulated use cases share the same coordination layer.

Area	Reusable in Dagents	Still specific to this case
Data connections	Local checks, small summaries, and source-side run history; no central raw-data store	Connect beside each hospital source; align terms, units, patient groups, and local history
Run coordination	Reusable pipelines, small-message contracts, run records, agent registration, and deployment plans	Production-grade queues, retries, schedules, storage, and recovery
Model management	Reusable model-run, result, testing, and version patterns	Build and clinically test a separate model, warning limit, and monitoring plan for each condition
Deployment	Reusable container and deployment instructions	Protect secrets, sign updates, isolate sites, test small rollouts, and plan recovery
Evidence + controls	Reusable plans, run IDs, policy checks, and change history	Access rules, tamper-resistant logs, staff training, audits, and clinical oversight

For one fixed model at one site, Dagents may be more infrastructure than needed / 18

DECISION GUIDE

Where Dagents fits—and where other tools lead.

Use Dagents when one coordinating layer must span hospitals, models, and deployment targets.

Option	Why teams choose it	Limit / Dagents fit
Dagents	Source-local and global agents, small-message workflows, model runs, versioned plans, and deployment instructions	Avoids a required raw-data lake, but is less mature and still needs production hardening and healthcare connectors
Federated-learning tools	Train or test models across sites without moving all raw data	Stronger for this specialist job; can run inside a Dagents-managed workflow
Cloud ML platforms	Managed training, model catalog, online endpoints, and monitoring	Faster cloud setup; hospital integration and dependence on one cloud still remain
Workflow platforms	Schedule steps, retry failures, reuse results, and run work on Kubernetes	Mature workflows, but they do not define hospital ownership or clinical safety rules
Clinical point product	A ready-made, validated product for one exact care workflow	Fastest for that exact use; less flexible across diseases, models, and hospitals

Dagents can coordinate specialist tools while keeping detailed records at the source / 19

PILOT ROADMAP

Evidence—not dates—moves the pilot forward.

Each phase must pass its review before the next phase begins; dates alone are not enough.

Phase	What happens	Proof needed
1. Define	Agree who the warning is for, what it predicts, which data it uses, exclusions, and owners	Approved case definition and shared data description
2. Test past data	Run the local agent on historical records stored at each hospital; send only approved summaries	Data quality, warning accuracy across groups, and normal warning volume
3. Run silently	Use live data but do not show warnings to clinicians	Speed, reliability, warning volume, complete records, and recovery results
4. Limited pilot	Show warnings to clinicians under an approved plan	Workflow usefulness, workload, overrides, usability, and safety review
5. Expand	Compare small hospital summaries, prove rollback, then prepare one additional condition	Multi-hospital evidence plus a separately approved definition and safety plan for the next condition

Expand one condition at a time; no central raw-patient lake or automatic care action

EVALUATION

A useful pilot must work beyond the model.

Model accuracy alone is not enough. Each condition must also be safe, dependable, understandable, and controllable.

Useful warnings

How many true cases are caught, how many warnings are correct, and how much advance notice is given.

Safe across groups

Check whether any patient group receives worse performance or too many warnings.

Reliable in practice

Fresh local data, successful source-side runs, response time, recovery, and time to return to a safe version.

Proof of control

Complete local history plus approved cross-hospital summaries, model versions, access exceptions, and site differences.

Set separate measured baselines for each condition; this deck does not invent clinical outcomes / 21

RECOMMENDATION

Start with sepsis; expand one condition at a time.

Use one source-first coordination platform, but never treat a sepsis model as a general medical model.

- Run the first silent pilot for sepsis because it tests urgent, multi-hospital coordination.
- Reuse the local/global controls for stroke, heart attack, kidney injury, and breathing decline.
- Give every condition its own clinical owner, approved inputs, tested model, workflow, and warning limits.
- Keep detailed patient data and run evidence beside each hospital source for every condition.
- Advance only after separate safety, usefulness, privacy, security, and rollback evidence is complete.

GLOSSARY

Hospital and model terms used in this deck.

Plain working definitions for this presentation—not medical or legal advice.

Term	Plain meaning	Why it matters here
Risk score	A number showing how strongly a tested model sees one defined risk	Supports clinician review; it is not a diagnosis and cannot cover every disease
EHR	A hospital's digital patient record	Source of vital signs, lab results, medicines, and visit information
PHI	Health information that can identify a person	Stays at the source; not copied into a central warehouse or lake for this case
Clinical decision support	Software advice shown to a clinician	The clinician—not the software—makes the care decision
Past-data test / silent live test	Test old records / run on live data without showing warnings	Safer steps before a limited clinician-facing pilot

GLOSSARY

Conditions mentioned in this deck.

Plain working definitions for this presentation—not medical advice or a diagnosis guide.

Term	Plain meaning	Why it matters here
Sepsis	The body's extreme response to infection; it can damage organs quickly	First detailed case because delays matter and many hospital signals must be coordinated
Stroke	An emergency where blood flow in the brain is blocked or a blood vessel breaks	Needs fast recognition and a separate stroke-specific workflow and safety proof
Heart attack	Blood flow bringing oxygen to part of the heart suddenly becomes blocked	Needs fast review and a separate heart-specific model and workflow
Sudden kidney injury	Kidney function falls quickly over hours or days	Needs local trend monitoring and separate kidney-specific rules and testing
Serious breathing decline	The lungs cannot provide enough oxygen or remove enough carbon dioxide	Needs oxygen and breathing monitoring with its own workflow and safety review

GLOSSARY

Project and compliance terms.

These labels describe project roles and control frameworks—not automatic proof of safety or compliance.

Term	Plain meaning	Why it matters here
LMA / GMA	Local agent in one hospital / global agent coordinating hospitals	Separates patient-level work from minimal, approved cross-hospital coordination
Data warehouse / lake	A central store that copies data from many systems	Useful for some analytics, but not required for this time-sensitive source-first design
HIPAA	U.S. rules for protecting patient health information	Guides privacy and security controls; architecture alone is not compliance
SOC 2	Independent review of service security and reliability controls	Requires working controls and evidence, not only design documents
Small rollout / rollback	Try an update at a few sites / return to the earlier safe version	Limits harm when a new model or setting behaves badly